



Sales Analysis Dashboard

Data Analysis using Power BI

Abdallah Mohamed Draz

Project Overview

This comprehensive sales analysis dashboard provides actionable insights into business performance across multiple dimensions. The goal is to enable data-driven decision-making by visualizing key metrics, identifying trends, and uncovering opportunities for growth.



Dataset Overview

The analysis leverages a comprehensive dataset capturing critical business metrics across multiple dimensions.



Sales Amount

Total revenue generated from transactions



Customers

Unique customer identifiers and demographics



Quantity

Number of units sold per transaction



Date

Year, quarter, month, and day granularity



Category

Product or service classification



Gender

Customer gender segmentation

Data Preparation

01

Data Cleaning

Removed duplicates and corrected inconsistencies in source data

02

Missing Values

Handled null values through appropriate imputation methods

03

Date Hierarchy

Built comprehensive time intelligence structure for analysis

04

Data Types

Converted fields to appropriate formats for calculations



Tools Used



Power BI

Primary visualization and reporting platform for interactive dashboards

Power Query

Data transformation and preparation tool for ETL processes

DAX

Data Analysis Expressions for advanced calculations and metrics

Key KPIs

150K

Total Sales

Revenue across all categories and time periods

850

Total Customers

Unique customers in the dataset

25K

Total Quantity

Units sold overall



Dashboard Overview

-  **Dynamic KPI Tracking**
Real-time monitoring of key metrics with trend analysis over time
-  **Interactive Filters**
Drill down by category, gender, and date ranges for focused analysis
-  **Visual Storytelling**
Charts and graphs that clearly communicate insights at a glance

Time Analysis



- **Year-over-Year Growth**

Track annual performance trends

- **Quarterly Patterns**

Identify seasonal fluctuations

- **Monthly Trends**

Analyze monthly performance

- **Daily Insights**

Spot day-of-week patterns

Category & Customer Analysis

Category Performance

Identified best-performing product categories and opportunities for improvement. Analysis revealed clear winners and underperformers.



Customer Insights

Gender-based analysis uncovered distinct purchasing patterns and distribution across customer segments.





Key Insights & Recommendations



Trends & Patterns

Consistent growth patterns and seasonal trends identified



Optimize Categories

Focus on high-performing categories, improve underperformers



Target Customers

Develop strategies for key customer segments

Conclusion

This dashboard demonstrates the power of data analytics in driving business decisions. By tracking KPIs, analyzing trends, and understanding customer behavior, organizations can optimize performance and increase revenue.